

The Self-Consistent D -Composite Scheme at Linear Order: W_L Subsector, Symbol Identity, Real-Frequency No-Go, and the Six-Dimensional Exponential Sector

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26 July 2026

Abstract

We audit the last unanalyzed variational scheme of the minimal branch: the self-consistent D -composite system in which the connection sits inside the defining jet. Exact results are obtained at the linearized, frozen-coefficient level around the affine point. (i) Imposing $\Theta \in W_L$ defines a consistent Lorentz-real subsector; necessity for self-consistent solutions is *not* proved, and a constant Hermitian shift at a constant tetrad is an explicit counterexample. (ii) The frozen symbol $A(s, \lambda)$ satisfies $A^3 = (\lambda \cdot s)A^2$ and $\text{tr } A^k = 6(\lambda \cdot s)^k$ for $k = 1, 2, 3$. Its characteristic polynomial is $t^{10}(t - q)^6$ with $q = \lambda \cdot s$ and $\det(I - A) = (1 - q)^6$. Generically the generalized zero-eigenspace contains three size-two and four size-one Jordan blocks. (iii) Off $q = 1$ the unique driven real-exponential solution is the exact gradient and is pullback-flat. For every real Fourier covector k , $q = i\lambda \cdot k$ and the determinant never vanishes, so this frozen torsion-free W_L linearization has no nonzero homogeneous curved real-frequency mode. (iv) At $q = 1$ there is an exactly six-dimensional real-exponential eigenspace. Its curl map is injective, and at several exact generic points its linearized Riemann map has rank six. Exceptional loci, variable-coefficient or nonlinear assembly, the quadratic action, gauge and ghost content, on-shell torsion, and the chain-rule spin current remain open. All algebraic statements are verified in `tools/verify_dcomposite_linearized.py`.

1 The self-consistent system

The scheme couples the covariant jet back to the connection:

$$E_\mu = \mathcal{N}_0^{-1/2} (\partial_\mu \Theta + \Omega_\mu[E] \Theta + \Theta \Omega_\mu[E]^\dagger), \quad g_{\mu\nu} = \frac{1}{2} (E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu), \quad (1)$$

with $\Omega_\mu[E]$ the spin representation of the Levi-Civita connection of the tetrad defined by the slice coordinates of E_μ (torsion-free branch first; anholonomy, not physical on-shell torsion, is the requirement for curvature).

Lemma 1 (W_L subsector). *Demanding $E_\mu \in W_L$ for every independently chosen Lorentz connection forces the Hermitian part of Θ to vanish; hence $\Theta \in W_L$ defines a consistent Lorentz-real subsector (exact solve; check L0). This does not prove necessity for self-consistent solutions, where only $\Omega = \Omega[E]$ occurs: $\Theta = \Theta_{\text{aff}} + H$ with constant Hermitian H has $\Omega[E] = 0$ and $D_\mu \Theta \in W_L$ although $\Theta \notin W_L$ (check L0b). All results below are statements about the W_L subsector.*

2 Linearization and the frozen symbol

The affine field $\Theta_{\text{aff}} = \Theta_0 + \sqrt{\mathcal{N}_0} \bar{E}_\nu x^\nu$ with $\Theta_0 \in W_L$ is an exact solution of (1) with $\Omega = 0$. Linearize $\Theta = \Theta_{\text{aff}} + \varepsilon \theta$, $E = \bar{E} + \varepsilon F$, freeze coefficients at x_0 , and pass to the real-exponential (Laplace-type) symbol $\partial_\nu \rightarrow s_\nu$ with real s . The system becomes

$$(I - A(s, \lambda)) F = \partial \theta', \quad \lambda^a = \text{slice coords of } \Theta_{\text{aff}}(x_0) / \sqrt{\mathcal{N}_0} = x_0^a + \theta_0^a, \quad (2)$$

where A implements the linearized Levi-Civita connection of the perturbed tetrad acting on the background through the bimodule. $\mathcal{N}_0$ cancels; A is bilinear in (s, λ) . Under the diagonal Lorentz group at the identity-tetrad background the mixed contraction $q = \lambda^a s_a$ is invariant.

Theorem 1 (Symbol identity, spectrum, and Jordan structure). *$A^3 = qA^2$ holds as a polynomial identity in all eight variables (check D2), and $\text{tr } A^k = 6q^k$ for $k = 1, 2, 3$ (check D2b). Hence the characteristic polynomial is $t^{10}(t - q)^6$ and*

$$\det(I - A) = (1 - q)^6. \quad (3)$$

At generic points $\text{rank } A = 9$ and $\text{rank } A^2 = 6$ (check D3), so the minimal polynomial is $t^2(t - q)$ and the generalized zero-eigenspace contains three size-two plus four size-one Jordan blocks. The frozen full symbol is modewise invertible iff $q \neq 1$; local or global uniqueness for the variable-coefficient PDE does not follow from this alone.

Corollary 1 (Frozen real-frequency no-go). *For a real Fourier covector k , substitute $s = ik$. Then $q = i\lambda \cdot k$ and*

$$\det(I - A(ik, \lambda)) = (1 - i\lambda \cdot k)^6 \neq 0, \quad (4)$$

because $|1 - ir|^2 = 1 + r^2 > 0$ for every real r . The homogeneous frozen symbol therefore has only the zero solution. Since A annihilates the exact-gradient image as a polynomial identity (check D1b), every driven real-frequency response is holonomic and linearly pullback-flat. Thus this frozen torsion-free W_L linearization contains no propagating anholonomic or curved Fourier mode (check D6).

Theorem 2 (Off- $q = 1$ flatness). *The linearized Levi-Civita connection factors through the curl of the tetrad perturbation, so A annihilates every exact gradient as a polynomial identity (checks D1–D1b). Consequently, for $q \neq 1$ the unique $\delta\Theta$ -driven solution of $(I - A)F = \partial\theta'$ is $F = \partial\theta'$ itself: exactly holonomic, with pullback-flat linearized geometry (check D4).*

Theorem 3 (Six-dimensional real-exponential sector). *At $q = 1$ the eigenspace $\ker(I - A)$ is exactly six-dimensional ($= \dim \mathfrak{so}(1, 3)$). The curl map is injective: a curl-free symbol mode is an exact gradient and is therefore annihilated by A , whereas an eigenmode obeys $AF = F$; hence a curl-free eigenmode must vanish. Exact multipoint checks D5–D5b confirm rank six. All linearized anholonomy of the frozen D -composite scheme is confined to this real-exponential sector.*

Corollary 2 (Where linearized curvature can live). *Any linearized curvature of the frozen scheme must be assembled from real-exponential symbol modes on the moving surface $(x_0 + \theta_0)s = 1$. At several exact generic points the restricted linearized Riemann map has rank six (check D5c). This proves a generic nonzero curvature image, not a theorem excluding exceptional loci. The sector is not a real-frequency propagating sector by Corollary 1. Its assembly into a variable-coefficient or nonlinear curved solution remains open.*

3 Answers to the six audit questions at this level

(1) Modewise invertibility of the frozen full real-exponential symbol: yes iff $q \neq 1$; PDE-level uniqueness remains open. For real Fourier covectors it is invertible for every k . (2) Anholonomy: driven solutions are exactly holonomic; anholonomy exists only in the real-exponential $q = 1$ sector. (3) $R \neq 0$ without inserting curvature by hand: the frozen real-frequency sector is a no-go; a generic nonzero linearized Riemann image exists only in the real-exponential sector, whose assembly into curved variable-coefficient solutions is open. (4) Chain-rule spin current: not yet computed. (5) On-shell torsion selection: not yet computed. (6) Degeneracy: outside the exponential $q = 1$ sector the scheme degenerates to the flat driven response. The six-dimensional

remainder is the only place where the quadratic action can yield nontrivial linearized geometric dynamics in this subsector.

Scope and non-claims

Frozen-coefficient analysis at the affine point, restricted to the W_L torsion-free subsector. It is not a solution theory for the variable-coefficient system: λ moves with x_0 , so the real-exponential condition sweeps spacetime and constant-coefficient solvability statements do not globalize automatically. Nothing here derives the Hilbert–Palatini term, decides on-shell torsion, or establishes curved variable-coefficient solutions. The real-frequency no-go applies only to this frozen torsion-free W_L linearization; nonlinear dynamics, variable coefficients, torsionful closures, and sectors outside W_L remain logically open. A discovered-and-fixed index transposition in the linearized connection is documented in the verifier as regression check D1; results computed with the faulty formula (the earlier $\{0^4, q^9, (q/2)^3\}$ spectrum) are void.